

Enterprise RAG Architecture Guide

This document explains the basics of Retrieval-Augmented Generation (RAG), vector databases, embeddings, chunking, semantic retrieval, and AI agents. The content is intentionally structured so it can be used for your first RAG implementation project.

What is RAG?

RAG stands for Retrieval-Augmented Generation. It combines a retrieval system with a Large Language Model. The retriever fetches relevant information from a knowledge base, and the LLM uses that context to generate grounded answers.

What are Embeddings?

Embeddings are vector representations of semantic meaning. Similar meanings are placed close together in vector space. Embedding models convert text into numerical coordinates that can be searched mathematically.

What is Chunking?

Chunking breaks large documents into smaller pieces. Each chunk should ideally contain one coherent concept. Good chunking improves retrieval quality and embedding relevance.

What is a Vector Database?

A vector database stores embeddings and enables similarity search. Popular vector databases include ChromaDB, Pinecone, Weaviate, and Qdrant.

What is Semantic Search?

Semantic search retrieves content based on meaning rather than exact keywords. This enables systems to understand concepts like 'authentication failure' and 'login issue' as related topics.

How Does Retrieval Work?

The user query is converted into an embedding. The vector database compares the query embedding with stored chunk embeddings and returns the nearest matches.

Prompt Grounding

Retrieved chunks are injected into the prompt context so the LLM can answer using trusted information rather than hallucinating.

What Makes an AI Agent?

An AI agent combines reasoning, memory, tools, and workflows. A basic RAG system is not yet a full agent because it does not autonomously plan or execute actions.

Component	Purpose
PDF Loader	Extract document text
Chunking	Create semantic units
Embeddings	Generate vector representations
ChromaDB	Store vectors and chunks
Retriever	Find relevant chunks
LLM	Generate final answer